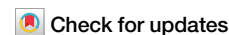


<https://doi.org/10.1038/s43247-025-02843-8>

Volcanically driven Hg–C–S cycling and climate change across the Early Permian Artinskian warming

Anbin Wu¹, Jian Cao¹ , Jingkun Zhang², Wenxuan Hu¹, Yuce Wang³, Chenjia Zhang¹ & Wenjun He⁴

Climate change and driving mechanisms of the Late Paleozoic Ice Age, an ideal analogue to the present-day global change, remain controversial. Here we propose a volcanic carbon–sulfur radiative forcing mechanism by investigating the Lower Permian Lucaogou Formation in the Junggar Basin, northern Pangea, across the Artinskian warming. Carbon isotope stratigraphy and paleoclimatic proxies record Early Permian carbon cycle fluctuations and second-order cooling amid first-order Artinskian warming. Peaks in mercury concentrations, mercury/total organic carbon, and negative $\delta^{13}\text{C}_{\text{org}}$ and $\delta^{34}\text{S}_{\text{bulk}}$ excursions align with Tarim Large Igneous Province eruptions, indicating that Tarim volcanism drove mercury–carbon–sulfur cycling and climate change. Specifically, sulfur cycling induced short-term cooling via elevated atmospheric sulfate, while carbon cycling fueled long-term warming by raising atmospheric carbon dioxide levels. Globally, volcanism likely modulated carbon–sulfur cycling and glacial–nonglacial alternations across geological eras, with sulfate forcing being a long-overlooked yet crucial mechanism for climate evolution.

The Late Paleozoic Ice Age (LPIA; main phase at *ca.* 330–254.5 Ma)^{1,2} archives the longest Phanerozoic icehouse and documents the only known climatic transition from a persistent icehouse to a greenhouse on Earth since the emergence of higher plants and complex terrestrial ecosystems^{3–7}. The LPIA is also the only geologic period where atmospheric CO₂ concentrations (*p*CO₂) most closely approximate those of the Quaternary glaciations^{8–10}, and thus is an important deep-time analogue for investigating present and future global environmental changes and has been highlighted for research for decades.

However, the driving mechanisms of the evolution of the LPIA are controversial, which hinders a comprehensive understanding of this event. Some studies have suggested that the atmospheric *p*CO₂ values during the LPIA were strongly coupled with surface temperatures and global glacial records^{11,12} and therefore attributed the evolution of the LPIA to carbon (C) cycle anomalies caused by atmospheric *p*CO₂ changes^{3,4,13,14}. The mainstream view is that the terrestrial C cycle was affected by the expansion of terrestrial higher plants, silicate weathering¹⁵, and changes in atmospheric *p*CO₂ caused by the Pangea Orogeny¹⁶. Nonetheless, the reconstructed atmospheric *p*CO₂ values do not ideally match the glacial record⁷. For instance, atmospheric *p*CO₂ exhibits an ambiguous relationship to the timing of onset, demise, and peak of the LPIA⁷. During 307–293.5 Ma, glaciers expanded continuously while *p*CO₂ fluctuated rather than

decreasing steadily¹⁷. Near the peak of this icehouse period (*ca.* 299–290 Ma), *p*CO₂ oscillated frequently both above and below the ~560 ppmv CO₂ glaciation threshold^{11,16,17}. As such, there appears to be some decoupling between glaciation and atmospheric *p*CO₂¹⁸.

Therefore, the evolution of the LPIA cannot be entirely attributed to atmospheric *p*CO₂, and other factors might be involved. Volcanic activity is widely recognized to have played a key role in the evolution of the LPIA, based on the timing of volcanic eruptions and climate–ice sheet models¹¹. In particular, during the Early Permian, large-scale volcanism (e.g., the Tarim large igneous province [LIP]) likely led to an increase in surface temperatures that marked the end of the Late Paleozoic glacial maximum^{12,19}. However, in terms of volcanic–climatic effects, besides causing warming, volcanism might also have injected large quantities of sulfur (S; e.g., ash, SO₂, and H₂S) into the stratosphere or troposphere, which were oxidized and reacted with the hydroxyl radical and H₂O to form sulfate aerosols²⁰. This would induce climatic cooling by increasing the planetary albedo, which could possibly initiate glacial events. For instance, the Late Permian Emeishan LIP eruptions released ~10¹⁷ g sulfur gases into the atmosphere, contributing to climate cooling during the Capitanian–Wuchiapingian transition (*ca.* 260 Ma)²¹ and thereby potentially triggering the P4 glaciation of the LPIA^{22,23}. Therefore, consideration of the radiative forcing effects of sulfate is essential to fully understand paleoclimate change and glacial

¹State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, School of Earth Sciences and Engineering, Nanjing University, Nanjing, China. ²State Key Laboratory of Palaeobiology and Stratigraphy, Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences, Nanjing, China. ³PetroChina Hangzhou Research Institute of Geology, Hangzhou, China. ⁴PetroChina Xinjiang Oilfield Company, Karamay, China. ✉e-mail: jcao@nju.edu.cn

events. However, the connection between LIP-driven C–S cycling and climatic changes and discrete glacial evolution during the LPIA remains poorly understood, thereby hampering understanding of its role in the evolution of the LPIA.

In this context, we investigated high-resolution continental stratum (the Lucaogou Formation, Junggar Basin) deposited from ~294 to ~285 Ma across the Sakmarian–Artinskian transition (*ca.* 290 Ma) in the mid- to high-latitude region (~45 °N) of northern Pangea^{24–26} (Fig. 1; Supplementary Note 1). This stratum is characterized by organic-rich deposits, substantial carbon sequestration^{26,27}, and abundant volcanic materials (e.g., ash, volcaniclastic debris, and tuff^{27,28}), with its volcanic records integral to the broader Tarim LIP^{29–31}. The interval represents a key period, coinciding with multiple critical events: a climate transition from a long-lived icehouse (340–290 Ma)¹² to a postglacial greenhouse state^{8,32–34}, the end of the Late Paleozoic glacial maximum^{12,19}, and dramatic volcanic–tectonic activities (e.g., the eruption of the Tarim LIP^{35,36} and the closure of the Ural Seaway³⁷). These temporal and spatial associations thereby offer an ideal archive for investigating Early Permian volcanism (or LIPs) and associated climatic–environmental perturbations. Here, we examined the links between volcanism, Sakmarian–Artinskian climate change, and the P1–P2 glacial history²² using sediment Hg concentrations and C–S isotope stratigraphy, to refine understanding of major geological events during the

LPIA and obtain a deep-time perspective on current and future global warming.

Results and discussion

Terrestrial response to the transition from Sakmarian icehouse to Artinskian greenhouse

We investigate if the Lucaogou Formation of the Junggar Basin preserves a terrestrial record of the Early Permian climatic transition, as observed in many strata globally, such as South Africa's Karoo Basin³⁸, the central United States and Russia³⁹, and the Yangtze Platform of South China⁴⁰. This can be realized by correlations of carbon isotope excursions (CIEs), which are widely used for global correlations of major geological events, such as extinctions, anoxic events, and climate change, based on the principle that carbon isotopes record carbon cycle perturbations generated by geological events^{41,42}.

In this study, the isotopic compositions of organic C ($\delta^{13}\text{C}_{\text{org}}$) and inorganic C ($\delta^{13}\text{C}_{\text{carb}}$) in the Lucaogou Formation of the Junggar Basin are -31.08‰ to -16.52‰ (average = -27.49‰) and -1.92‰ to 12.71‰ (average = 7.22‰), respectively (Supplementary Note 2). Compared with the uniform values in the Lower Member, the $\delta^{13}\text{C}_{\text{org}}$ and $\delta^{13}\text{C}_{\text{carb}}$ values in the Upper Member are more variable, exhibiting synchronous but generally opposing trends, and can be divided into intervals I, II, and III (Fig. 2a, b). These CIEs, including positive inorganic CIEs (*p*CIEs) and negative organic CIEs (*n*CIEs), can be employed to reflect environmental and climatic

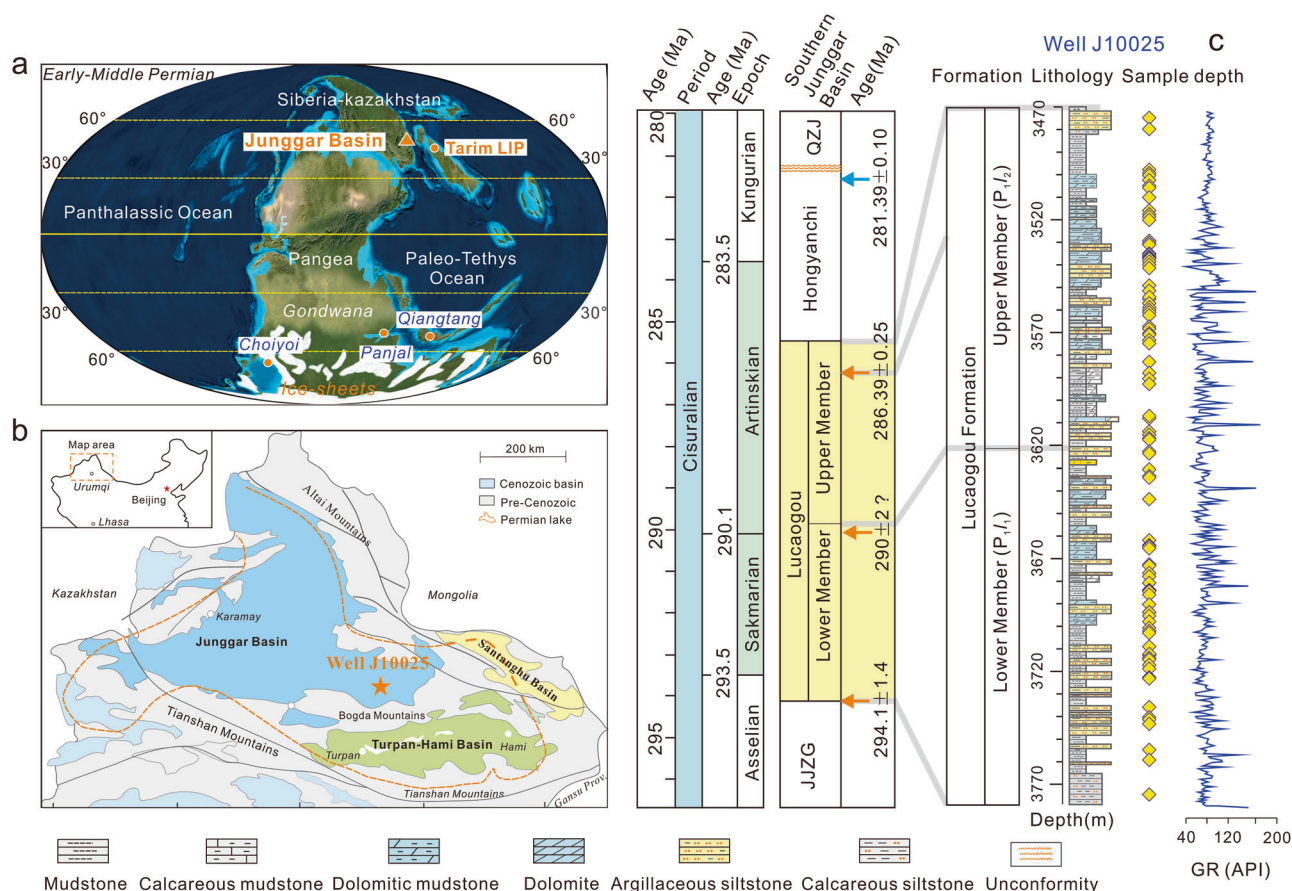


Fig. 1 | Geological setting of the Junggar Basin and sampling locations as well as stratigraphic framework of the Lucaogou Formation. **a** The location of the Junggar Basin and the distribution of some typical volcanoes, including Tarim LIP, Qiangtang, Panjal, and Choyoi, are shown in the Early-Middle Permian paleogeography (modified from refs. 89,90.). **b** Regional structural framework of the Junggar Basin²⁶, location of Well J10025 (orange star), and distribution of key

geological units (including the Turpan - Hami Basin, Santanghu Basin, and major mountain ranges) are shown. The maximum known extent of Permian lake deposits is delineated by orange dashed lines²⁶. **c** Chronological framework, stratigraphy^{23,24}, lithology, and sample depth (yellow rhombuses) of the Lucaogou Formation in Well J10025 are presented. JJZG and QZJ denote the Jingjingzigou and Quanzijie Formations.

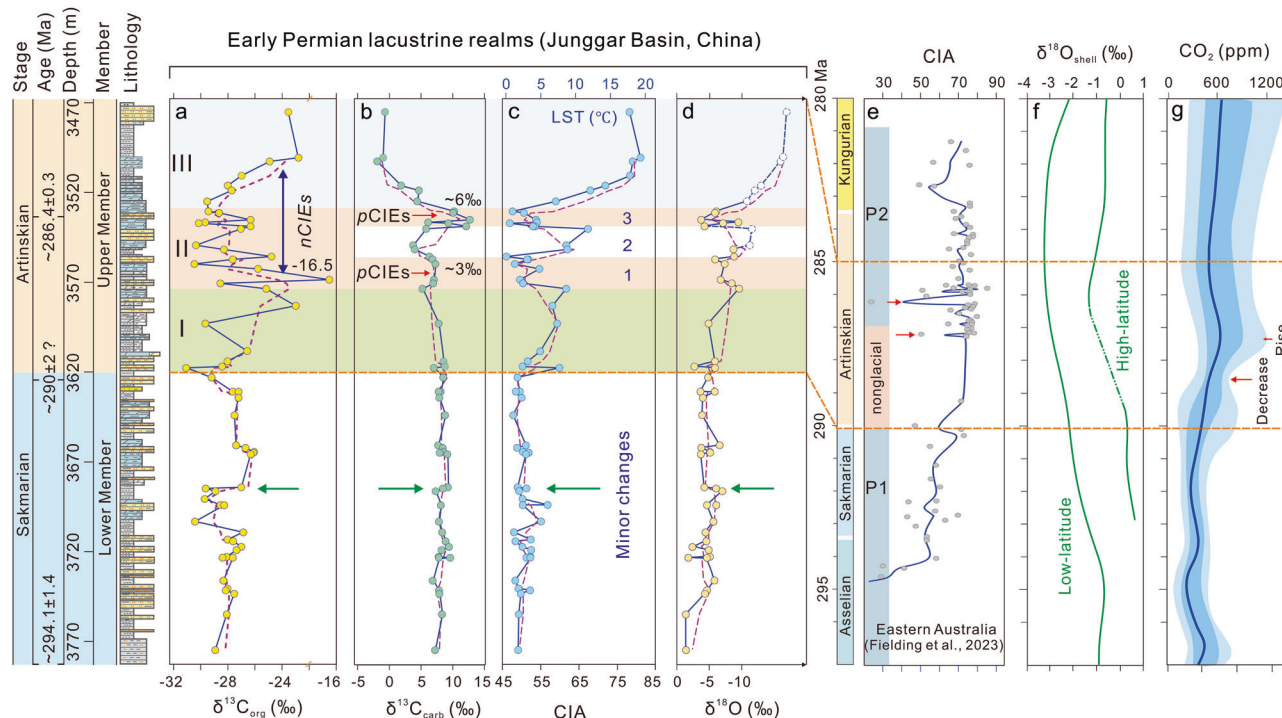


Fig. 2 | Change trends and differences of carbon-oxygen isotopes and CIA in the upper and lower members of the lucaogou formation (Early Permian, Junggar Basin), compared with those in global marine strata across the Artinskian warming event. a, b Stratigraphic changes in $\delta^{13}\text{C}_{\text{org}}$ and $\delta^{13}\text{C}_{\text{carb}}$ values (pCIEs and nCIEs refer to positive inorganic and negative organic carbon isotope excursions, respectively). c, d Stratigraphic changes in CIA and LSTs and $\delta^{18}\text{O}$ values (note that

the dashed circles represent data $<-10\%$; these are used primarily to help show the overall trend in oxygen isotopes rather than denoting the absolute value of the original fluid signal, with their trend indicated by the dashed line). e CIA change in the eastern Australia during the P1–P2 glaciations. f Changes in oxygen isotopic compositions of low-Mg calcite from low- and high-latitude fossil shells^{46,47}. g Early Permian global atmospheric $p\text{CO}_2$ variations¹².

changes, as they are not caused by diagenesis, organic matter degradation, enrichment, sources (and mixing), thermal evolution, and kerogen type (Supplementary Note 3.1). Based on chronostratigraphic constraints²⁴, the CIEs and $\delta^{13}\text{C}$ trends in the Lucaogou Formation of the Junggar Basin are consistent with global marine and terrestrial strata record during the Sakmarian–Artinskian, such as those in the Yuzhou section in North China⁴³, marine–coastal strata in the United States and Russia³⁹, Zhongdi section in South China⁴⁰ and the Dal’ny Tulkas section in southern Russia⁴⁴ (Supplementary Fig. 4). Thus, the CIEs in the Lucaogou Formation of the Junggar Basin reflect Artinskian carbon cycle perturbations, which may have been closely related to changes in atmospheric $p\text{CO}_2$ during the P2 stage of the LPIA^{11,45} (Supplementary Note 3.1).

The chemical index of alteration (CIA) and land surface temperature (LSTs) were further used to examine paleoclimatic changes (Supplementary Note 3.2) to assess whether the climatic changes recorded by the carbon cycle in the Lucaogou Formation of the Junggar Basin are consistent with the global record. Results show that the CIA values of the Lucaogou Formation increase from 45–55 in the Lower Member to 55–85 in the Upper Member, and the corresponding LSTs also increase from 0–5 °C to 5–20 °C (Fig. 2c), which is consistent with the Sakmarian–Artinskian warming recorded by the increase in CIA values in the Karoo Basin of South Africa³⁸ and eastern Australia (Fig. 2e)². Given the sensitivity of carbonate oxygen isotopic compositions to temperature⁴⁴, the trend is also indicative of a temperature change. The $\delta^{18}\text{O}$ values of the Lucaogou Formation decrease by $> 5\%$ from the Lower to Upper Member (Fig. 2d), which is consistent with the Sakmarian–Artinskian warming recorded by the continuous decrease in $\delta^{18}\text{O}$ values of marine low-Mg calcite fossil shells from high- and low-latitude regions of the Northern Hemisphere (Fig. 2f)^{46,47}. Two cooling events (II-1 and –3) were further identified in interval II, which exhibits the largest carbon isotope fluctuations, corresponding to negative CIA excursions (Fig. 2c), positive $\delta^{18}\text{O}$ excursions (Fig. 2d), and a decrease in

LSTs of ~ 6 °C (Fig. 2e). This is also evidenced by other paleoclimatic indicators, such as C values, Fe/Mn, (Al + Fe)/(Mg + Ca), Rb/Sr and Sr/Cu ratios, which also show synergistic changes with LSTs (CIA) and $\delta^{18}\text{O}$ (Supplementary Fig. 2 and Supplementary Note 3.2). The two short-term cooling events were superimposed on a warming trend.

The climatic changes and discrete glacial–nonglacial cycles recorded in the eastern Australia (e.g., C1–C3 and P1–P4) are the most detailed Late Paleozoic glacial–stratigraphic framework currently available^{2,22} and thus serve as an ideal analogue for global comparisons of the LPIA. The sedimentary and geochemical responses to the Late Paleozoic glacial events have also been obtained for several other basins (e.g., in south Africa, Brazil, and Antarctica)². Both the Lucaogou Formation and the formation in eastern Australia exhibit an increasing trend in CIA values, with lower values in the P1 stage, a gradual increase in the nonglacial interval of P1–P2 stages, and negative excursions in the initial P2 stage (Fig. 2e). This indicates that the Lucaogou Formation of the Junggar Basin likely records the climatic fluctuations of the P1–P2 discrete glacial–nonglacial events (Supplementary Fig. 6), which may have been related to Early Permian atmospheric $p\text{CO}_2$ changes, as the LSTs curve from the Lucaogou Formation is similar to the Sakmarian–Artinskian atmospheric $p\text{CO}_2$ curve (Fig. 2c, g)¹².

In summary, geochemical data from the Lucaogou Formation in the Junggar Basin document Artinskian carbon cycle perturbations and the second-order cooling events within the Artinskian warming stages. This represents a identified terrestrial record of the Early Permian climate evolution in the Northern Hemisphere, with implications of global significance.

Early Permian volcanism and Hg–C–S cycles

In the Lucaogou Formation of this study, the sediment mercury (Hg) concentrations, total organic carbon (TOC), total sulfur (TS) contents, and TS/TOC ratios in the nCIEs intervals are elevated and associated with negative $\delta^{34}\text{S}_{\text{bulk}}$ (bulk sulfur isotope) excursions (nSIEs; amplitude $> 10\%$; Fig. 3a–c), collectively indicative of coupled Hg–C–S cycling and associated

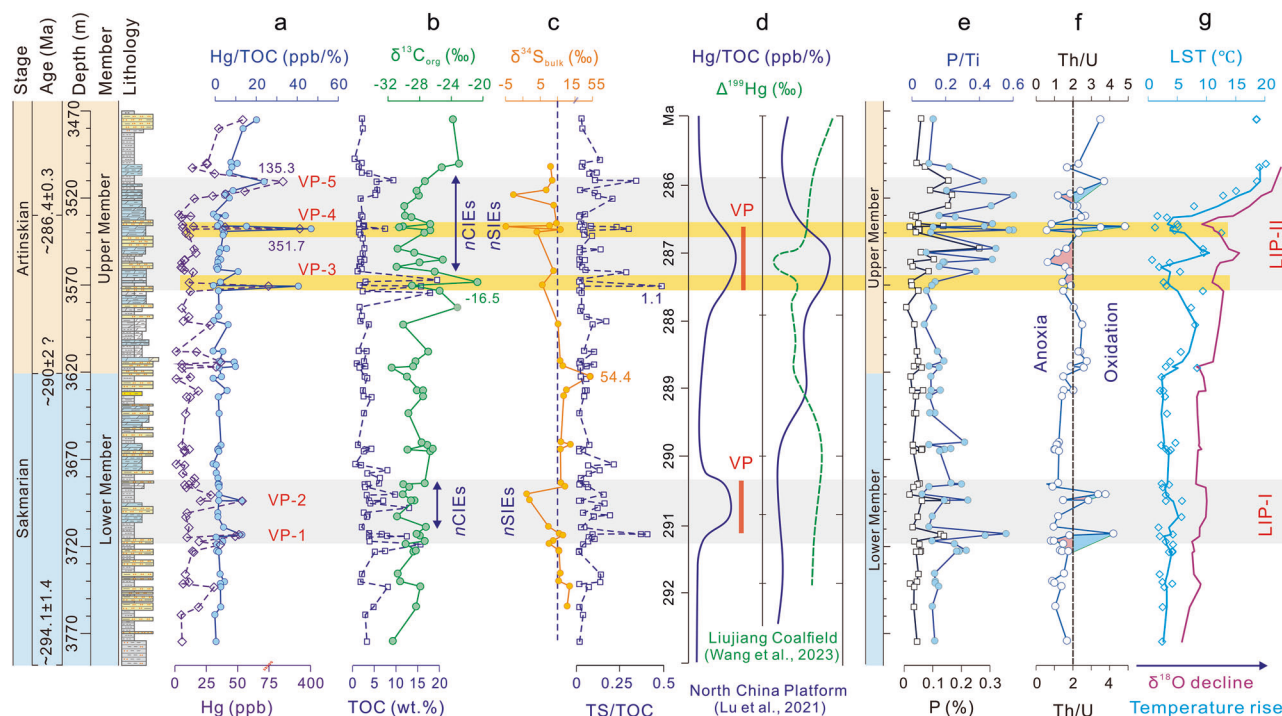


Fig. 3 | Variations in volcanic activity, Hg–C–S geochemical data (concentration and isotopes), and other related parameters in the Sakmarian–Artinskian Lucaogou Formation of the Junggar Basin. **a** Covariation of sediment Hg concentrations and Hg/TOC ratios. The peak values in Hg/TOC corresponds to that in Hg concentrations. Five volcanic events of different intensities (VP-1, –2, –3, –4, and –5) in two stages (LIP-I and -II) were identified (light gray shaded area).

b Covariation of TOC contents and $\delta^{13}\text{C}_{\text{org}}$ values. negative $\delta^{13}\text{C}_{\text{org}}$ excursions (nCIEs) occur synchronously with increasing TOC contents. **c** Covariation of TS/

TOC ratios and $\delta^{34}\text{S}_{\text{bulk}}$ (bulk sulfur isotope) values. Negative $\delta^{34}\text{S}_{\text{bulk}}$ excursions (nSIEs) occur synchronously with increasing TS/TOC values. **d** Hg/TOC ratios in isochronous strata of the North China Platform and $\Delta^{199}\text{Hg}$ values, which together record volcanic activity^{43,57}. **e** Covariation between a nutrient element (P) and geochemical proxy of primary productivity (P/Ti). **f** Variations in redox state (Th/U). **g** Covariation of land surface temperatures (LSTs) and $\delta^{18}\text{O}$ values, and their relationship to volcanic activity.

climate change. Sediment Hg enrichment and concentration anomalies are generally associated with volcanism and are reliable fingerprints of past volcanic activity^{48–50}, and thus sediment Hg concentrations were used to assess whether volcanism was the main driver of the Sakmarian–Artinskian C–S cycle fluctuations (Supplementary Note 3.3).

The background Hg concentrations of the Lucaogou Formation are 0–20 ppb, and there are five anomalies with high Hg concentrations in two stages (LIP-I and -II; Fig. 3a), with values of 50.1–351.7 ppb (average = 132.7 ppb) (Supplementary Data 1 and Note 2). The adsorbed form of Hg in the Lucaogou Formation is mainly onto organic matter (Supplementary Note 3.3), and thus Hg/TOC ratios were used in this study to trace volcanism and its intensity (Fig. 3a). The background Hg/TOC ratios are 0–5 ppb/% and the anomalies represent by high Hg/TOC values of 13.9–46.9 ppb/% (average = 28.0 ppb/%). The five Hg/TOC anomalies further include two peaks in the P1 glacial (LIP-I; average Hg/TOC = 14.0 ppb/%, represented by VP-1 and –2) and three larger peaks in the P2 glacial (LIP-II; average Hg/TOC = 37.4 ppb/%, represented by VP-3, –4, and –5) (Fig. 3a). Compared with published global Hg concentration data affected by volcanic activity (mainly at the Permian–Triassic [P–T] and Triassic–Jurassic [T–J] transitions, and in the Early Jurassic), with Hg concentration peaks of 50–400 ppb^{51–54}, the data for the Lucaogou Formation of this study are similar. Typically, volcanic eruptions lead to an increase in Hg loading in the atmosphere–land–ocean system and modify Hg–TOC systematics^{49,51,55}, which is also consistent with the synchronous increase in Hg concentrations and TOC contents in the Lucaogou Formation (Fig. 3a, b). Meanwhile, The Hg/TOC ratios peaks are synchronous with nCIEs (Fig. 3a, b), implying that the Hg enrichment in the Lucaogou Formation was the result of volcanism and organic-matter-rich sediments were affected by thermocatalysis during the ascent and intrusion of magma^{54,56}, thereby inducing nCIEs.

Based on eruption timing and spatial distributions^{29–31}, the two volcanic stages (LIP-I and LIP-II) recorded in the Lower Permian Lucaogou Formation of this study, which are genetically linked to the Tarim LIP^{35,36}, align with Hg/TOC ratios and $\Delta^{199}\text{Hg}$ anomalies in coeval North China Platform strata (Fig. 3d)^{43,57}. This fine correlation indicates that Tarim LIP volcanism extended beyond the Tarim and Junggar basins (Fig. 1a), reaching wider regions (e.g., the North China Craton), thereby driving widespread nCIEs across these areas^{43,57}. This coincided with Artinskian carbon cycle perturbations in global marine–terrestrial strata. Overall, these findings suggest that large-scale volcanism was likely the primary driver of Sakmarian–Artinskian carbon cycle perturbations and warming.

In addition to the volcanically-driven Hg–C cycle as discussed above, volcanism also controlled the sulfur cycle during the deposition of the Lucaogou Formation by regulating changes in lake sulfate concentrations, as evidenced by the coupled elevated TS/TOC ratios and nSIEs in the volcanic layers (Fig. 3c and Supplementary Note 3.4). The large amount of sulfur (e.g., H_2S and SO_2) released by volcanic eruptions reacted with hydroxyl radical in the atmosphere and was rapidly converted into H_2SO_4 ²⁰. The H_2SO_4 condensed into droplets on pre-existing particles or nucleated to produce new particles, which formed sulfate aerosols²⁰ and eventually settled to the Earth surface system as acid rain and increased the sulfate concentration in the Lucaogou Formation (i.e., elevated TS/TOC ratios; Fig. 3c). This resulted in the weakening of microbial sulfate reduction (MSR) and the sulfate reservoir effect⁵⁸, which caused the nSIEs (Fig. 3c) in association with the high Hg concentrations due to volcanism (Fig. 3a). Therefore, the nSIEs in the Lucaogou Formation record perturbations of the sulfur cycle caused by the Early Permian volcanism (e.g., Tarim LIP). Given the global significance of the carbon isotope stratigraphy of the Lucaogou Formation as discussed above, the correlated changes in the C–S cycle suggest that

volcanic activity was likely also an additional driver of Early Permian global sulfur cycle perturbations.

In summary, Early Permian large-scale volcanism (e.g., the Tarim LIP) elevated atmospheric CO₂ and SO₂ concentrations, driving synchronized fluctuations in the Hg–C–S cycles.

Volcanically forced climate change

Based on the above results and discussion, a model of the C–S cycle is established to explain the short- and long-term climatic effects across the Artinskian warming triggered by volcanism represented by the Lucaogou Formation of the Junggar Basin (Fig. 4).

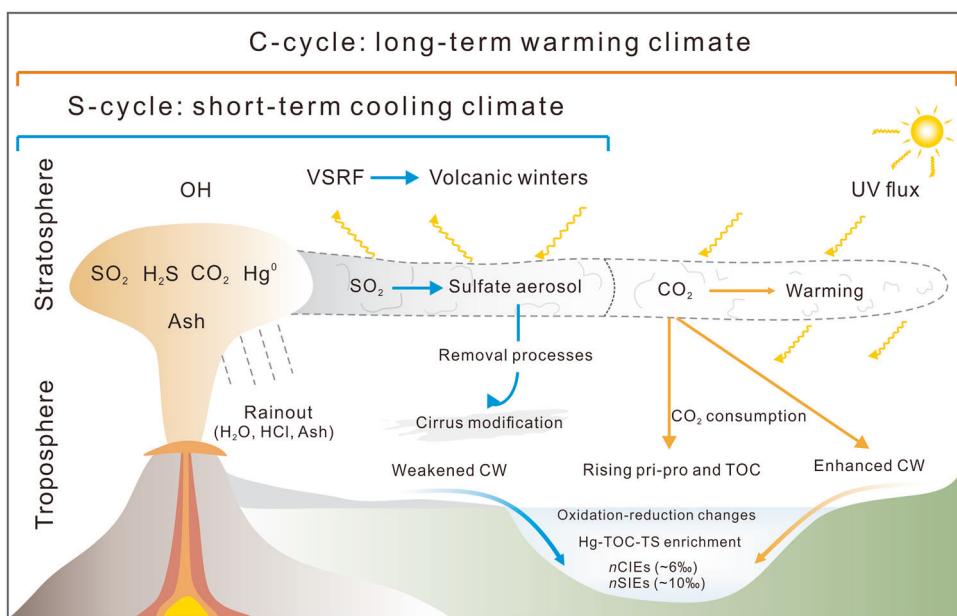
Firstly, the volcanic sulfate radiative forcing (VSRF) caused short-term climatic cooling and may contribute to intermittent glacial periods (Fig. 4). Volcanic activity drives synergistic changes in the Hg–C–S cycle, with a consistent response across multiple sedimentary basins, especially in the carbon cycle (Fig. 5a–d). The timing of the LIP eruption recorded in the Lucaogou Formation coincides with the initiation of the P2 discrete glacial (Fig. 5a, b), and the rapid decrease in LST values and increase in oxygen isotopes (Fig. 3g), as well as the synchronized *n*SIEs (Fig. 3c), suggest a rapid decrease in ambient temperatures at the onset of the eruption, rather than a sustained warming, which is mainly attributed to the VSRF effect. Based on the depositional duration (over ca. 3 Myr) and sedimentation rate (8–10 cm kyr^{−1}) of the Lucaogou Formation during the greenhouse period²⁶, two volcanic cooling events during the LIP-II with temperature drops of ~6.6 °C and ~9.3 °C are roughly estimated to have cumulative durations of ~93.5 and ~167.5 kyr, respectively (Fig. 3a, g). Because the volcanism resulted in a marked increase in atmospheric sulfate concentration, the sulfate aerosols reflected ultraviolet rays and scattered solar radiation back to space, which cooled Earth's surface^{20,59}. Subsequent acid rain resulted in the high TS concentrations and TS/TOC ratios, and *n*SIEs in the Lucaogou Formation (Fig. 3c).

However, due to the weaker volcanic intensity (low Hg peaks: ~50 ppb) coupled with generally lower temperatures (0–5 °C) during the LIP-I, the cooling effect of VSRF was relatively insignificant, and thus no notable fluctuations in CIA and LSTs values occurred within this interval (Fig. 3g). In addition, the volcanism also enhanced the primary productivity (high P/Ti ratios; Fig. 3e) and organic carbon burial (high TOC contents; Fig. 3b) by rapidly injecting large amounts of nutrients (e.g., elevated P contents; Fig. 3e) into the Earth's surface (or lake) system, further contributing to *n*CIEs, increasing the consumption and sequestration of atmospheric CO₂, and amplifying cooling effects.

In contrast to the sulfur cycle, the volcanic carbon cycle causes long-term climatic warming (Fig. 4) and likely drives a shift from glacial to nonglacial conditions. During the transition from icehouse to greenhouse conditions, silicate weathering in the Lucaogou Formation intensified gradually (Fig. 2c), and organic carbon was buried (Fig. 3b), which is consistent with the global organic carbon enrichment in Lower Permian sedimentary rocks⁶⁰. Theoretically, these processes drawdown and consume large amounts of CO₂ from the atmosphere–ocean system, which reduces atmospheric *p*CO₂ and cools the climate, which is inconsistent with the persistent climatic warming of the Early Permian. This suggests that there were other exogenous factors compensating for the reduction of atmospheric *p*CO₂ and led to warming. This was most likely due to volcanism. Based on evidence from the Lucaogou Formation of this study, the volcanism during the glacial stage (i.e., during deposition of the Lower Member of the formation) was small in scale and of low intensity (sediment Hg concentration ~ 50 ppb), which caused small perturbations in the carbon cycle (*n*CIEs < 1.5‰; Fig. 3a, b) and almost no climate change (Fig. 3g). Volcanism during the greenhouse stage (i.e., during deposition of the Upper Member) was much larger in scale and intensity (sediment Hg concentrations up to ~350 ppb), which caused carbon cycle perturbations (*n*CIEs: ~6‰; Fig. 3a, b) and had dramatic climatic effects (Fig. 3g).

To further quantify the volcanic–climatic effects, the carbon input fluxes were roughly estimated based on the cGENIE model (a complex Earth system model⁶¹). Using a sedimentation rate of 8–10 cm kyr^{−1} as a baseline²⁶, the duration of the multi-stage volcanism during the glacial (Lower Member) and greenhouse (Upper Member) periods of the Lucaogou Formation were calculated to be 104.5 and 291.5 kyr, respectively (with corresponding *n*CIEs of ~1.5‰ and ~6.5‰, respectively; Fig. 3a, b). If the input carbon source is only mantle-derived C (δ¹³C = −6‰), then during the glacial period, the total carbon input and maximum annual carbon flux into the paleo-Junggar Lake from volcanism were 2–6 × 10³ Pg C and 0.1–0.2 Pg C yr^{−1}, respectively. During the greenhouse period, these values were 0.5–1.0 × 10⁵ Pg C and 3–4 Pg C yr^{−1}, respectively. The highest CO₂ emissions during the Artinskian period are roughly half those of the Siberian LIPs (~2.0 × 10⁵ Pg C)³⁵. This difference arises mainly because, relative to the peak eruption rates (8.0 km³ yr^{−1}) and volumes (3–4 × 10⁶ km³) of Siberian LIPs, the Tarim LIP is relatively smaller in scale, with corresponding values of ~0.06 km³ yr^{−1} and 0.15–0.30 × 10⁶ km³, respectively^{35,36}. However, The highest flux of volcanic input (0.5–1.0 × 10⁵ Pg C and 3–4 Pg C yr^{−1}) clearly exceed both the total organic carbon burial flux (10⁴ Pg C) and maximum carbon burial rate

Fig. 4 | Effects of volcanic C–S cycle on Early Permian climate change. Early Permian volcanism (e.g., Tarim LIP) that drove C–S cycling and associated long-term climatic warming and short-term climatic cooling are recorded in the Early Permian large-scale paleo-Junggar lake, Northern Pangea (modified from ref. 91 and references therein). VSRF, UV and OH refer to volcanic sulfate radiative forcing, ultraviolet and hydroxyl radical, respectively. CW, Pri-pro, *n*CIEs and *n*SIEs refer to chemical weathering, primary productivity, negative organic carbon and bulk sulfur isotope excursions, respectively.



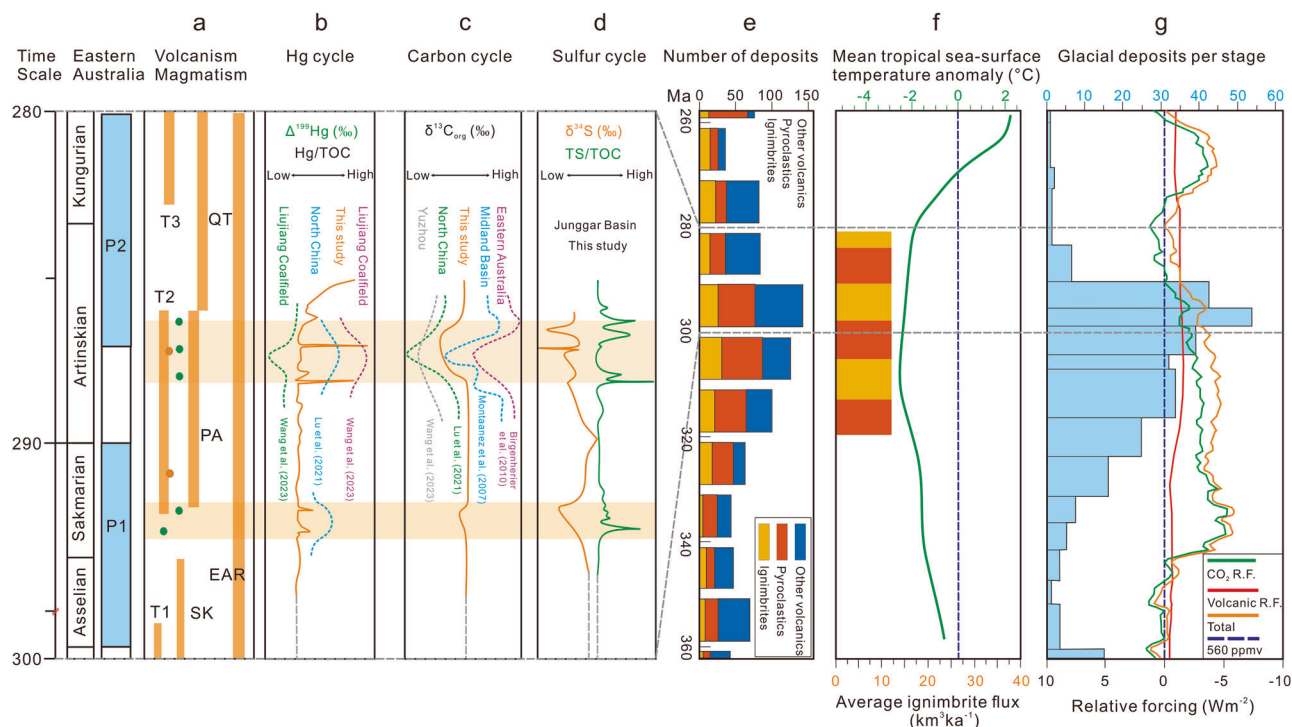


Fig. 5 | Coupling of volcanism and its C–S radiative forcing effects with climate change and glacial evolution during the LPIA. a Early Permian spatiotemporal framework, showing timing constraints on P1–P2 glaciations in East Australia⁷ and volcanic and magmatic activities during 300–280 Ma (or P1–P2 interval). EAR (East Australian Rift⁶⁷); T1–T3 (first, second and third phases of Tarim LIP³⁶); SK (Skagerrak–Centred LIP⁷¹); PA (Panjal Traps LIP⁶⁵); QT (Qiantang Basalts⁶⁶). Orange^{43,57} and blue dots (this study) represent volcanic peaks tracked by Hg/TOC during the corresponding periods. (b–d) Early Permian Hg–C–S cycle, synchronous with the Tarim LIP and other volcanism. **b** Sharp increase in Hg/TOC ratios accompanied by negative $\Delta^{199}\text{Hg}$ excursion in volcanic interlayers of the Junggar Basin and North China Plateau. **c** Variable magnitude $n\text{CIEs}$ in global marine–terrestrial strata during the Artinskian warming, coinciding with LIP

eruption phases. **d** Sharp increase in TS/TOC ratios with $n\text{SIEs}$ in the volcanic interlayers of Junggar Basin. **e** Histogram of volcanics by type, binned at 10 m.y. intervals. “Other volcanics” refers to volcanic products excluding pyroclastics or ignimbrites as coded in original sources⁷². **f** Silicic LIPs record (red), modified from Bryan (2007)⁷⁸, representing silicic volcanic provinces with documented volumes $>10^5 \text{ km}^3$. Mean tropical sea-surface temperature (green line) has been detrended and smoothed using a 50 Ma window with 10 Ma steps⁷³. **g** Radiative forcing from volcanic sulfate and CO_2 (ca. 360–260 Ma) versus glacial deposits per stage (note: the glacial deposition data herein excludes the updates from ref. 2). Comparison of negative radiative forcing attributable to $p\text{CO}_2$ (green line), volcanic sulfate aerosol (red line), and total radiative forcing (orange line) relative to 560 ppmv CO_2 (dashed line)⁷².

($4 \times 10^{-3} \text{ Pg C yr}^{-1}$) of the Lucaogou Formation²⁶. This implies that volcanic carbon sources released by Tarim LIP may have mixed with other isotopically depleted carbon sources (e.g., via thermocatalytic release of terrestrial or rock-bound organic carbon or microbial methane release from hydrate reservoirs)^{61,62}, yielding an input $\delta^{13}\text{C}$ of -12‰ to -6‰ ⁶¹ and thereby triggering the $n\text{CIEs}$ in the Lucaogou Formation.

For the volcanically release of SO_2 , we present a rough estimation based on magma volume. The Emeishan LIP has been extensively investigated, which has a total magma volume exceeding $3.0 \times 10^5 \text{ km}^3$ and at least 1500 Gt of SO_2 was released^{35,63}. For comparison, the total magma volume of the Tarim LIP is estimated to $\sim 1.5\text{--}3.0 \times 10^5 \text{ km}^3$ ^{36,64}, and thus it may have released more than 750 Gt of SO_2 . Volcanism at the periphery of the Tarim LIP could have resulted in the released SO_2 flux being $>1500 \text{ Gt}$ with twice the total eruptive area of the LIP at $\sim 6 \times 10^5 \text{ km}^2$ ^{29–31}. These eruptions, combined with other large global volcanic events (e.g., the Panjal Traps⁶⁵, Qiantang dike⁶⁶; Fig. 5a) in the Early Permian, may have caused intermittent volcanic winters and the discrete glacial events. Therefore, the C–S radiative forcing revealed in this study provides an explanatory perspective on the evolution of the LPIA.

Linkage between volcanism and the evolution of discrete glaciations during the LPIA

The LPIA is composed of eight discrete glacial–nonglacial intervals (Carboniferous C1–C4; Permian P1–P4), where intervals preserving glacial activity and cold-climate indicators alternated with those lacking such evidence^{2,22}. Notably, these cyclically repeated glaciations are generally

correlated with widespread intraplate magmatic events^{65–68}, including the multipulsed (327–250 Ma) European–northwest African magmatic province (EUNWA; and its initiation as the 300 Ma Skagerrak Centered LIP (SLIP)), and the 260 Ma Emeishan LIP of China (Fig. 5a)^{69–71}. However, how volcanism shaped the evolution of discrete glaciations and the LPIA remains unclear sufficiently.

The volcanic C–S radiative forcing mechanism elucidated herein offers a perspective on the evolution of LPIA, and the VSRF may have contributed to initiate and sustain the icehouse conditions. From the perspective of LPIA evolution, during the peak of the volcanic eruption (ca. 300–290 Ma), the frequency of glacial deposit, average ignimbrite flux, and anomalies in tropical mean sea-surface temperature all reached their maxima (Fig. 5e, f)^{72,73}. In contrast, during periods when the frequency and intensity of eruptions declined (e.g., after 290 Ma and before 320 Ma), both sea-surface temperature anomalies and glacial deposit volumes diminished (Fig. 5e, f). All these patterns thus reflect cooling effect potentially linked to the VSRF mechanism, as supported by the C–S radiative forcing model (Fig. 5g)⁷². Volcanic sulfate forcing maintains effective $p\text{CO}_2$ at a level near or below the 560 ppmv glaciation threshold from ca. 360 to 257 Ma, corresponding exactly to the interval recording glacial deposits of the LPIA (Fig. 5g). The interval of maximum frequency of glacial deposits (ca. 298–295 Ma) falls within maximum (negative) radiative forcing (-1.6 Wm^{-2}) induced by sulfate aerosols (Fig. 5g)⁷². Thus, sulfate aerosols likely counteracted Milankovitch-scale and longer-period $p\text{CO}_2$ maxima, sustaining long-term icehouse conditions from the Late Carboniferous into the Permian⁷².

Beyond VSRF, volcanism induced sulfate anomalies and elevated carbon burial rates, which also contribute to low $p\text{CO}_2$ levels. Firstly, volcanic materials are rich in nutrient (e.g., Si, Fe, and P) that can enhance marine and terrestrial primary productivity by stimulating biological proliferation. This nutrient-driven process efficiently sequesters atmospheric CO_2 . Secondly, volcanism elevates sulfate concentrations, increasing the acidity in marine–terrestrial environments and promoting dissolution of reactive Fe⁷⁴. The Fe fertilization effect further boosts primary production and carbon sequestration. Notably, elevated P contents, P/Ti ratios (Fig. 3e) and TOC contents (Fig. 3b) in the volcanic layers, alongside high Fe contents in the Permian Lucaogou Formation, offer solid evidence for these processes.

Conversely, as volcanism ceased, nutrient supply to the lake declined. This led to a decrease in primary productivity (P/Ti; Fig. 3e) and TOC contents (Fig. 3b), while promoting the accumulation and emission of greenhouse gases (e.g., CO_2) that drive climate warming. Meanwhile, rising temperatures will exacerbate the oxidation of the terrestrial–marine environment and accelerate the CH_4 cycling, an ecosystem process highly temperature-dependent^{75,76}. These patterns are also evident in the Lucaogou Formation. From icehouse (Lower Member) to greenhouse period (Upper Member), the Lucaogou Formation shows a decline in average TOC contents (4.6% to 3.6%; Supplementary Data 1) and thermal maturity (vitrinite reflectance, $R_o = 0.84$ to 0.65)⁷⁷, alongside shifts toward more oxic depositional environments (Th/U ratios = 1.5 to 2.1; Supplementary Data 1) and intensified methanogenesis²⁵. This indicates a decrease in carbon sequestration and an increase in carbon emissions from icehouse to greenhouse period in the paleo-Junggar Lake, aligning with the sharp decline in Early Permian global terrestrial burial fluxes of organic carbon (peaking at ~ 300 Ma)⁶⁰. Collectively, these changes in the carbon cycle intensified climate warming and favored the shift from icehouse to greenhouse conditions.

In conclusion, while this study offers a perspective on the cooling effects caused by LIP eruptions, it should be noted that not all LIPs have the capacity to affect climate change. Instead, silicic supereruptions—particularly those yielding felsic to intermediate magma compositions—are the primary drivers of sulfur injection into the stratosphere^{72,78}. The cooling effects are harder to document than long-term warming effects, as the former are easily overprinted or superimposed by the latter. Exploring the links between volcanism, magmatism, and glacial evolution thus remains a key challenge, requiring more high-precision data and research. Additionally, glacial evolution is likely driven by multiple interacting factors, including tectonic activity and tectonic belt drift^{34,79}, orogeny¹⁶, silicate and ophiolite weathering^{12,15}, and terrestrial carbon cycles (e.g., vegetation-induced carbon sequestration)^{3,14}. These factors cannot be ignored, and their integrated effects merit further exploration.

Our findings also have implications for the modern geoengineering initiative to mitigate global warming, one of whose potential measures is to counteract warming by injecting sulfate aerosols into the stratosphere⁸⁰, a process similar to the VSRF of this study. Based on our results, sulfate aerosols do exert short-term cooling effects but cannot prevent long-term warming and may adversely affect the environment (e.g., acidification, altered redox state; Fig. 3c, f). Thus, in a comprehensive perspective, carbon emission reductions, rather than VSRF, represent a more effective long-term strategy for slowing global warming and sustaining a habitable Earth.

Conclusions

Geochemical proxies from the Lucaogou Formation in the Junggar Basin record Artinskian warming events and second-order climatic fluctuations (especially cooling) across the discrete P1–P2 glacial intervals of the LPIA. These changes are associated with the Tarim-LIP-driven multi-stage volcanism, which substantially perturbed the Hg–C–S cycles. The volcanic sulfur cycle elevated atmospheric sulfate concentrations, cooling the climate and potentially triggering the discrete P2 glaciation. However, the VSRF effect cannot prevent the persistent Artinskian warming, while the volcanic carbon cycle reinforced long-term warming.

Based on the Sakmarian–Artinskian record in the Junggar Basin, the dual radiative forcing effect of the LIPs-driven C–S cycling needs to be fully considered when explaining the short- and long-term climatic changes, and glacial–nonglacial cycles. Incorporating negative VSRF effect into atmospheric $p\text{CO}_2$ models improves consistency with the glacial evolution of the LPIA. The VSRF effect likely represents a long-overlooked yet crucial mechanism for initiating and sustaining glacial evolution of the LPIA, and thereby triggers a series of carbon cycle-related events, which likely aid in maintaining low atmospheric $p\text{CO}_2$ levels during the LPIA. However, owing to the scarcity of high-precision data and paleoclimate records, further investigation into volcanic carbon and sulfur cycling during the LPIA is required to systematically bridge the knowledge gap. This might be general for the critical events throughout geological history.

Methods

Sampling

The samples were collected from the Lucaogou Formation in well J10025 in the Jimusar Sag, Junggar Basin (Fig. 1b, c). Ninety-three samples from the Lower and Upper Members were used for sedimentological and geochemical investigations. The samples were made into thin-sections for petrographic observations and powdered to 200 mesh for geochemical analysis. The total organic C, and total N and S (TOC, TN, and TS, respectively) contents, major and trace element concentrations, and (in) organic C and O isotopic compositions were determined at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, Nanjing University, Nanjing, China. Sulfur isotope analysis was undertaken at the International Center for Isotope Effects Research, Nanjing University and sedimentary mercury (Hg) concentrations were determined at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Wuhan, China.

Element mapping

Element distributions in smooth rock surfaces and thin-sections were semi-quantitatively determined with an X-ray fluorescence (XRF) spectrometer element scanner. The instrument consists of a 50 μm micro-focusing source, convergent X-ray lens, and Mo target, and was operated at a maximum voltage of 50 kV, maximum current of 1 mA, continuously adjustable power, and scanning interval of 13–20 μm . The relative content of each element from Al to Ba was obtained in counts per second (cps).

Mercury concentrations

We weighed 80–150 mg of dried sample powder (80–120 mg for mudstone or shale and 120–150 mg for limestone) into a Ni capsule for analysis with a LECO® AMA254 Hg analyzer (the Ni capsule was first oven-dried at a high temperature to minimize the Hg blank). To ensure the data were accurate, a standard and duplicate sample were analyzed after every 13 samples. The coal standard 502–685 yielded a Hg content of 41 ± 8 ppb, with a relative standard deviation (RSD) of less than $\pm 5\%$.

Total organic carbon, total nitrogen, and total sulfur contents

The TOC, TN, and TS contents were determined with a Vario Macro CHNS elemental analyzer (ECS 4024 CHNSO; Costech). Firstly, 10% HCl was reacted with ~ 2 g of sample powder for 48 h to completely remove carbonates, and then the solution was centrifuged at 3000 rpm for 10 min and the solid residue was washed with deionized water. This procedure was repeated 7–8 times until the solution pH reached 7. Finally, the residual solid was dried at 60 °C, and 30 mg of the sample was wrapped in Sn foil for analysis. The accuracy of the data is better than $\pm 0.1\%$.

Major- and trace- elements analyses

The powdered samples were dried in an oven to remove volatiles, and then ~ 1 g of powdered sample was mixed with ~ 11 g of lithium tetraborate, melted at 1050 °C, and cooled to form glass disks for analysis. The samples were analyzed for major elements using a Thermo Scientific ARL 9900 XRF spectrometer. The voltage and current were 40 kV and 70 mA, respectively,

and the scan time for each element was 20 s. Based on the measured values for the standards (BHVO-2 and BCR-2), the relative error was $\pm 1\%$ when the element abundance was >1.0 wt.% and $\pm 10\%$ when the element abundance was <1.0 wt.%. An Element XR inductively coupled plasma–mass spectrometer (ICP–MS) was used to determine the trace element contents. An aliquot of each sample (50.0 ± 0.2 mg) was dissolved in a mixture of HF and HNO₃, and digested in high-pressure Teflon bombs. Rhodium was used as an internal standard and the data accuracy was $<5\%$.

Organic carbon isotopes

Organic C isotopes ($\delta^{13}\text{C}_{\text{org}}$) were determined with a Finnigan MAT 253 mass spectrometer. The powdered sample was acidified and air-dried, mixed with CuO powder (mass ratio of 1:8) in a quartz tube that was vacuum-sealed, and heated to 850°C . After cooling in a cryogenic trap to separate H₂O, CO₂ was directly introduced into the inlet system of the mass spectrometer. The standard was GBW04407 ($\delta^{13}\text{C}_{\text{V-PDB}} = -22.43\text{‰} \pm 0.07\text{‰}$), and the results are reported relative to Vienna Pee Dee Belemnite (V-PDB) with an accuracy of $\pm 0.1\text{‰}$.

Inorganic carbon and oxygen isotopes

The C and O isotopic compositions of whole-rock samples were determined with a Thermo Finnigan Delta V Plus continuous flow isotope ratio mass spectrometer (IRMS). An aliquot of 80–120 μg of sample powder was dissolved in orthophosphoric acid at 70°C for >12 h and connected to the continuous flow sample preparation device (Gas Bench II) of the IRMS. V-PDB was used as the standard. All isotope ratios are reported in per mil (‰), and the C and O isotope data accuracy is better than $\pm 0.1\text{‰}$ and $\pm 0.5\text{‰}$, respectively.

Sulfur isotopes

For the whole-rock sulfur isotopes analyses. Firstly, 10% HCl was reacted with dried sample powders (<200 mesh) at 25°C to completely remove carbonates, and then the solution was centrifuged at 3000 rpm for 10 min and the solid residue was washed with >18 M $\Omega\cdot\text{cm}^{-1}$ deionized water. This procedure was repeated 7–8 times until the solution pH reached ~ 7 . Finally, the residual solid was dried at 60°C for sulfur isotope analysis. Sulfur isotopes were analyzed with a MAT253 Plus mass spectrometer and Flash EA elemental analyzer coupled with a Conflo IV multi-purpose interface instrument. The pre-treated samples were combusted at high temperatures in an O₂-permeable environment, reduced and converted to SO₂ in high-purity He carrier gas, and passed through a redox reaction tube to fully oxidize all the gases. Finally, the SO₂ gas was separated using a chromatographic column and analyzed by mass spectrometry. The S isotope results are expressed in standard delta notation as per mil (‰) relative to the Vienna Canyon Diablo Troilite (V-CDT) standard, with an analytical accuracy of $\pm 0.2\text{‰}$.

$$\delta^{34}\text{S}(\text{‰}) = ((^{34}\text{S}/^{32}\text{S})_{\text{sample}} / (^{34}\text{S}/^{32}\text{S})_{\text{standard}} - 1) \times 1000 \quad (1)$$

Chemical index of alteration and land surface temperatures

The chemical index of alteration (CIA) is widely used to assess deep-time changes in climate and chemical weathering, and is calculated as follows⁸¹:

$$\text{CIA} = [\text{Al}_2\text{O}_3 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O} + \text{K}_2\text{O})] \times 100 \quad (2)$$

where CaO* is the CaO bound in silicates, excluding the CaO in carbonates or apatite⁸², and $\text{CaO}_{\text{excess}} = \text{CaO} - \text{P}_2\text{O}_5 \times 10/3$. If $\text{CaO}_{\text{excess}} < \text{Na}_2\text{O}$, then $\text{CaO}^* = \text{CaO}_{\text{excess}}$; otherwise, $\text{CaO}^* = \text{Na}_2\text{O}$ ⁸³.

The chemical index of weathering (CIW) and modified CIA (CIX) can correct for the effects of K and Ca on the CIA parameters. The CIW is used to remove K₂O from the CIA to avoid the effect of K metasomatism⁸⁴, and the CIX is used to remove CaO* from CIA to avoid the effects of

carbonates⁸⁵. The formulae are as follows:

$$\text{CIW} = [\text{Al}_2\text{O}_3 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O})] \times 100 \quad (3)$$

$$\text{CIX} = \text{Al}_2\text{O}_3 / (\text{Al}_2\text{O}_3 + \text{Na}_2\text{O} + \text{K}_2\text{O}) \times 100 \quad (4)$$

Furthermore, the CIA values of modern suspended particulate matter in large global rivers are sensitive to land surface temperatures (LSTs), estuary latitude, and soil depth in the watershed⁸⁶. As such, CIA values can be used to estimate LSTs using the following quantitative relationship³³:

$$\text{LST}(\text{°C}) = 0.56 \times \text{CIA} - 25.7 (R^2 = 0.50) \quad (5)$$

which is applicable at temperatures of $3\text{--}25^\circ\text{C}$, corresponding to a CIA range of approximately 50–90. This LSTs estimation has been successfully applied to quantify paleotemperatures of Permian and Triassic terrestrial successions^{20,33,87}.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data is available in the main text or the supplementary information and the data table has been deposited in Figshare⁸⁸ (Wu Anbin & Cao Jian, <https://doi.org/10.6084/m9.figshare.28942166.v2>).

Received: 23 May 2025; Accepted: 26 September 2025;

Published online: 11 November 2025

References

1. Saltzman, M. R. Late Paleozoic ice age: Oceanic gateway or $p\text{CO}_2$? *Geology* **31**, 151–154 (2003).
2. Fielding, C. R., Frank, T. D. & Birgenheier, L. P. A revised, late Palaeozoic glacial time-space framework for eastern Australia, and comparisons with other regions and events. *Earth Sci. Rev.* **236**, 104263 (2023).
3. Berner, R. A. *The Phanerozoic Carbon Cycle: CO₂ and O₂*. Oxford University Press (2004).
4. McKenzie, N. R. et al. Continental arc volcanism as the principal driver of icehouse-greenhouse variability. *Science* **352**, 444–447 (2016).
5. Isaacson, P. E., Díaz-Martínez, E., Grader, G. W., Babek, O. & Devuyst, F. X. Late Devonian–earliest Mississippian glaciation in Gondwanaland and its biogeographic consequences. *Paleogeogr. Paleoclimatol. Paleoecol.* **268**, 126–142 (2008).
6. Lakin, J. A., Marshall, J. E. A., Troth, I. & Harding, I. C. Greenhouse to icehouse: a biostratigraphic review of latest Devonian–Mississippian glaciations and their global effects. *Geol. Soc. Spec. Publ.* **423**, 439–464 (2016).
7. Foster, G. L., Royer, D. L. & Lunt, D. J. Future climate forcing potentially without precedent in the last 420 million years. *Nat. Commun.* **8**, 14845 (2017).
8. Montañez, I. P. et al. CO₂-forced climate and vegetation instability during late Paleozoic deglaciation. *Science* **315**, 87–91 (2007).
9. Montañez, I. P. et al. Climate, $p\text{CO}_2$ and terrestrial carbon cycle linkages during late Palaeozoic glacial–interglacial cycles. *Nat. Geosci.* **9**, 824–828 (2016).
10. Berner, R. A. GEOCARBSULF: A combined model for Phanerozoic atmospheric O₂ and CO₂. *Geochim. Cosmochim. Ac.* **70**, 5653–5664 (2006).
11. Montañez, I. P. Late Paleozoic climate window of opportunity. *Proc. Natl. Acad. Sci. USA* **113**, 2334–2336 (2016).
12. Richey, J. D. et al. Influence of temporally varying weatherability on CO₂-climate coupling and ecosystem change in the late Paleozoic. *Clim. Past* **16**, 1759–1775 (2020).

13. Jagoutz, O., Macdonald, F. A. & Royden, L. Low-latitude arc-continent collision as a driver for global cooling. *Proc. Natl. Acad. Sci. USA* **113**, 4935–4940 (2016).
14. Royer, D. L., Berner, R. A., Montañez, I. P., Tabor, N. J. & Beerling, D. J. CO₂ as a primary driver of Phanerozoic climate. *GSA today* **14**, 4–10 (2004).
15. Cleal, C. J. & Thomas, B. A. Palaeozoic tropical rainforests and their effect on global climates: Is the past the key to the present? *Geobiology* **3**, 13–31 (2005).
16. Goddérís, Y. et al. Onset and ending of the late Palaeozoic ice age triggered by tectonically paced rock weathering. *Nat. Geosci.* **10**, 382–386 (2017).
17. Montañez, I. P. Current synthesis of the penultimate icehouse and its imprint on the Upper Devonian through Permian stratigraphic record. *Geol. Soc. Spec. Publ.* **512**, 213–245 (2022).
18. Horton, D. E. & Poulsen, C. J. Paradox of late Paleozoic glacioeustasy. *Geology* **37**, 715–718 (2009).
19. Marchetti, L. et al. The Artinskian Warming Event: an Euramerican change in climate and the terrestrial biota during the early Permian. *Earth-Sci. Rev.* **226**, 103922 (2022).
20. Kremser, S. et al. Stratospheric aerosol-Observations, processes, and impact on climate. *Rev. Geophys.* **54**, 278–335 (2016).
21. Yang, J., Cawood, P. A. & Du, Y. Voluminous silicic eruptions during late Permian Emeishan igneous province and link to climate cooling. *Earth Plan. Sci. Lett.* **432**, 166–175 (2015).
22. Fielding, C. R. et al. Stratigraphic imprint of the late Palaeozoic Ice Age in eastern Australia: a record of alternating glacial and nonglacial climate regime. *J. Geol. Soc. London* **165**, 129–140 (2008).
23. Shellnutt, J. G. The Emeishan large igneous province: A synthesis. *Geosci. Front.* **5**, 369–394 (2014).
24. Yang, W. et al. Depositional environments and cyclo- and chronostratigraphy of uppermost Carboniferous–Lower Triassic fluvial–lacustrine deposits, southern Bogda Mountains, NW China—A terrestrial paleoclimatic record of mid-latitude NE Pangea. *Glob. Planet. Change* **73**, 15–113 (2010).
25. Sun, F. et al. Sustained and intensified lacustrine methane cycling during Early Permian climate warming. *Nat. Commun.* **13**, 4856 (2022).
26. Carroll, A. R. & Wartes, M. A. Organic carbon burial by large Permian lakes, northwest China. *Geol. Soc. Am. Spec. Pap.* **370**, 91–104 (2003).
27. Gao, Y. et al. Paleoenvironmental setting, mechanism and consequence of massive organic carbon burial in the Permian Junggar Basin, NW China. *J. Asian Earth Sci.* **194**, 104222 (2020).
28. Wu, H. G. et al. A unique lacustrine mixed dolomitic–clastic sequence for tight oil reservoir within the middle Permian Lucaogou Formation of the Junggar Basin, NW China: Reservoir characteristics and origin. *Mar. Pet. Geol.* **76**, 115–132 (2016).
29. Zhang, C. L. et al. A Permian large igneous province in Tarim and Central Asian orogenic belt, NW China: Results of a ca. 275 Ma mantle plume? *Geol. Soc. Am. Bull.* **122**, 2020–2040 (2010).
30. Qin, K. Z. et al. SIMS zircon U–Pb geochronology and Sr–Nd isotopes of Ni–Cu bearing mafic–ultramafic intrusions in eastern Tianshan and Beishan in correlation with flood basalts in Tarim basin (NW China): Constraints on a ca. 280 Ma mantle plume. *Am. J. Sci.* **311**, 237–260 (2011).
31. Yang, S. F. et al. Early Permian Tarim Large Igneous Province in northwest China. *Sci. China Earth Sci.* **56**, 2015–2026 (2013).
32. Montañez, I. P. & Poulsen, C. J. The Late Paleozoic ice age: an evolving paradigm. *Annu. Rev. Earth Planet. Sci.* **41**, 629–656 (2013).
33. Yang, J., Cawood, P. A., Du, Y., Feng, B. & Yan, J. Global continental weathering trends across the Early Permian glacial to postglacial transition: Correlating high- and low-paleolatitude sedimentary records. *Geology* **42**, 835–838 (2014).
34. Griffis, N. P. et al. Coupled stratigraphic and U–Pb zircon age constraints on the late Paleozoic icehouse-to-greenhouse turnover in south-central Gondwana. *Geology* **47**, 1146–1150 (2019).
35. Chen, J., & Xu, Y. G. Permian large igneous provinces and their paleoenvironmental effects. In *Large Igneous Provinces: A Driver of Global Environmental and Biotic Changes*, (eds Ernst, R. E., Dickson, A. J. & Bekker, A.) Vol. 255, 417–434 (Geophysical Monograph Series, AGU and Wiley, Hoboken, NJ, 2021).
36. Xu, Y. G., Wei, X., Luo, Z. Y., Liu, H. Q. & Cao, J. The early Permian Tarim large Igneous Province: Main characteristics and a plume incubation model. *Lithos* **204**, 20–35 (2014).
37. Kwon, H. et al. Responses of the biogeochemical sulfur cycle to Early Permian tectonic and climatic events. *Earth Plan. Sci. Lett.* **591**, 117604 (2022).
38. Scheffler, K., Hoernes, S. & Schwark, L. Global changes during Carboniferous–Permian glaciation of Gondwana: Linking polar and equatorial climate evolution by geochemical proxies. *Geology* **31**, 605–608 (2003).
39. Grossman, E. L. et al. Glaciation, aridification, and carbon sequestration in the Permo–Carboniferous: the isotopic record from low latitudes. *Paleogeogr. Paleoclimatol. Paleoecol.* **268**, 222–233 (2008).
40. Buggisch, W., Wang, X., Alekseev, A. S. & Joachimski, M. M. Carboniferous–Permian carbon isotope stratigraphy of successions from China (Yangtze platform), USA (Kansas) and Russia (Moscow Basin and Urals). *Paleogeogr. Paleoclimatol. Paleoecol.* **301**, 18–38 (2011).
41. Magioncalda, R., Dupuis, C., Smith, T., Steurbaut, E. & Gingerich, P. D. Paleocene–Eocene carbon isotope excursion in organic carbon and pedogenic carbonate: Direct comparison in a continental stratigraphic section. *Geology* **32**, 553–556 (2004).
42. Saltzman, M. R., Thomas, E. & Gradstein, F. M. Carbon isotope stratigraphy. *The geologic time scale* **1**, 207–232 (2012).
43. Lu, J. et al. Records of organic carbon isotopic composition ($\delta^{13}\text{C}_{\text{org}}$) and volcanism linked to changes in atmospheric $p\text{CO}_2$ and climate during the Late Paleozoic Icehouse. *Glob. Planet. Change* **207**, 103654 (2021).
44. Zeng, J., Cao, C. Q., Davydov, V. I. & Shen, S. Z. Carbon isotope chemostratigraphy and implications of palaeoclimatic changes during the Cisuralian (Early Permian) in the southern Urals, Russia. *Gond. Res.* **21**, 601–610 (2012).
45. Isbell, J. L. et al. Glacial paradoxes during the late Paleozoic ice age: Evaluating the equilibrium line altitude as a control on glaciation. *Gond. Res.* **22**, 1–19 (2012).
46. Korte, C., Jasper, T., Kozur, H. W. & Veizer, J. $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ of Permian brachiopods: A record of seawater evolution and continental glaciation. *Paleogeogr. Paleoclimatol. Paleoecol.* **224**, 333–351 (2005).
47. Korte, C., Jones, P. J., Brand, U., Mertmann, D. & Veizer, J. Oxygen isotope values from high-latitudes: Clues for Permian sea-surface temperature gradients and Late Palaeozoic deglaciation. *Paleogeogr. Paleoclimatol. Paleoecol.* **269**, 1–16 (2008).
48. Percival, L. M. et al. Does large igneous province volcanism always perturb the mercury cycle? Comparing the records of Oceanic Anoxic Event 2 and the end-Cretaceous to other Mesozoic events. *AM. J. Sci.* **318**, 799–860 (2018).
49. Grasby, S. E., Them, T. R., Chen, Z., Yin, R. & Ardakani, O. H. Mercury as a proxy for volcanic emissions in the geologic record. *Earth Sci. Rev.* **196**, 102880 (2019).
50. Shen, J. et al. Sedimentary host phases of mercury (Hg) and implications for use of Hg as a volcanic proxy. *Earth Planet. Sci. Lett.* **543**, 116333 (2020).
51. Grasby, S. E., Sanei, H., Beauchamp, B. & Chen, Z. Mercury deposition through the Permo–Triassic biotic crisis. *Chem. Geol.* **351**, 209–216 (2013).

52. Thibodeau, A. M. et al. Mercury anomalies and the timing of biotic recovery following the end-Triassic mass extinction. *Nat. Commun.* **7**, 1–8 (2016).
53. Percival, L. M. et al. Mercury evidence for pulsed volcanism during the end-Triassic mass extinction. *Proc. Natl. Acad. Sci. USA* **114**, 7929–7934 (2017).
54. Shen, J. et al. Mercury evidence of intense volcanic effects on land during the Permian-Triassic transition. *Geology* **47**, 1117–1121 (2019).
55. Sanei, H., Grasby, S. E. & Beauchamp, B. Latest Permian mercury anomalies. *Geology* **40**, 63–66 (2012).
56. Shen, J. et al. Negative C-isotope excursions at the Permian-Triassic boundary linked to volcanism. *Geology* **40**, 963–966 (2012).
57. Wang, Y. et al. Volcanism and wildfire associated with deep-time deglaciation during the Artinskian (early Permian). *Glob. Planet. Change* **225**, 104126 (2023).
58. Canfield, D. E. Biogeochemistry of sulfur isotopes. *Rev. Mineral Geochem.* **43**, 607–636 (2001).
59. Robock, A. Volcanic eruptions and climate. *Rev. Geophys.* **38**, 191–219 (2000).
60. Boyce, C. K., Ibarra, D. E. & D'Antonio, M. P. What we talk about when we talk about the long-term carbon cycle. *New Phytol.* **237**, 1550–1557 (2023).
61. Vervoort, P., Adloff, M., Greene, S. E. & Kirtland Turner, S. Negative carbon isotope excursions: an interpretive framework. *Environ. Res. Lett.* **14**, 085014 (2019).
62. Lyons, S. L. et al. Palaeocene–Eocene thermal maximum prolonged by fossil carbon oxidation. *Nat. Geosci.* **12**, 54–60 (2019).
63. Chen, J. & Xu, Y. G. Permian large igneous province and their impact on paleoenvironment and biodiversity: progress and perspectives. *Bull. Mineral. Petrol. Geochem.* **36**, 374–393 (2017).
64. Usui, Y. & Tian, W. Paleomagnetic directional groups and paleo-intensity from the flood basalt in the Tarim large igneous province: implications for eruption frequency. *Earth, Planets Space* **69**, 14 (2017).
65. Shellnut, J. G. Igneous rock associations 21. The Early Permian Panjal Traps of the Western Himalaya. *Geosci. Can.* **43**, 251–264 (2016).
66. Zhang, Y. X. & Zhang, K. J. Early Permian Qiangtang flood basalts, northern Tibet, China: A mantle plume that disintegrated northern Gondwana? *Gond. Res.* **44**, 96–108 (2017).
67. Korsch, R. J., Totterdell, J. M., Cathro, D. L. & Nicoll, M. G. Early Permian East Australian Rift System. *Aust. J. Earth Sci.* **56**, 381–400 (2009).
68. Youbi, N. et al. Preliminary appraisal of a correlation between glaciations and large igneous provinces over the past 720 million years. In *Large Igneous Provinces: A Driver of Global Environmental and Biotic Changes* (eds Ernst, R. E., Dickson, A. J. & Bekker, A.) Vol. 255 169–190 (Geophysical Monograph Series, AGU and Wiley, Hoboken, NJ, 2021).
69. Doblas, M. et al. Permo-Carboniferous volcanism in Europe and northwest Africa: A superplume exhaust valve in the center of Pangaea. *J. Afr. Earth. Sci.* **26**, 89–99 (1998).
70. Wilson, M. et al. Permo-Carboniferous magmatism and rifting in Europe. *Geol. Soc. Spec. Publ.* **223**, 1–10 (2004).
71. Torsvik, T. H., Smethurst, M. A., Burke, K. & Steinberger, B. Long term stability in deep mantle structure: evidence from the ~300 Ma Skagerrak - Centered large Igneous Province (the SCLIP). *Earth Plan. Sci. Lett.* **267**, 444–452 (2008).
72. Soreghan, G. S., Soreghan, M. J. & Heavens, N. G. Explosive volcanism as a key driver of the late Paleozoic ice age. *Geology* **47**, 600–604 (2019).
73. Cather, S. M. et al. Climate forcing by iron fertilization from repeated ignimbrite eruptions: The icehouse–silicic large igneous province (SLIP) hypothesis. *Geosphere* **5**, 315–324 (2009).
74. Oakes, M. et al. Iron solubility related to particle sulfur content in source emissions and ambient fine particles. *Environ. Sci. Technol.* **46**, 6637–6644 (2012).
75. Pancost, R. D. et al. Increased terrestrial methane cycling at the Palaeocene–Eocene thermal maximum. *Nature* **332**–335 (2007).
76. Yvon-Durocher, G. et al. Methane fluxes show consistent temperature dependence across microbial to ecosystem scales. *Nature* **507**, 488–491 (2014).
77. Zhang, R. et al. First report of Early Permian Tasmanite oil shales in northern Pangea with global implications. *Org. Geochem.* **188**, 104736 (2024).
78. Bryan, S. E. Silicic large igneous provinces. *Episodes* **30**, 20–31 (2007).
79. Limarino, C. O. et al. A paleoclimatic review of southern South America during the late Paleozoic: A record from icehouse to extreme greenhouse conditions. *Gond. Res.* **25**, 1396–1421 (2014).
80. Kravitz, B. et al. Climate model response from the geoengineering model intercomparison project (GeoMIP). *J. Geophys. Res.: Atmospheres* **118**, 8320–8332 (2013).
81. Nesbitt, H. & Young, G. M. Early Proterozoic climates and plate motions inferred from major element chemistry of lutites. *Nature* **299**, 715–717 (1982).
82. Fedo, C. M., Wayne Nesbitt, H. & Young, G. M. Unraveling the effects of potassium metasomatism in sedimentary rocks and paleosols, with implications for paleoweathering conditions and provenance. *Geology* **23**, 921–924 (1995).
83. McLennan, S. M. Weathering and global denudation. *J. Geol.* **101**, 295–303 (1993).
84. Harnois, L. The CIW index: A new chemical index of weathering. *Sediment Geol.* **55**, 319–322 (1988).
85. Garzanti, E., Padoan, M., Setti, M., López-Galindo, A. & Villa, I. M. Provenance versus weathering control on the composition of tropical river mud (southern Africa). *Chem. Geol.* **366**, 61–74 (2014).
86. Li, C. & Yang, S. Is chemical index of alteration (CIA) a reliable proxy for chemical weathering in global drainage basins? *Am. J. Sci.* **310**, 111–127 (2010).
87. Frank, T. D. et al. Pace, magnitude, and nature of terrestrial climate change through the end-Permian extinction in southeastern Gondwana. *Geology* **49**, 1089–1095 (2021).
88. Wu, A. B. & Cao, J. Volcanically driven Hg–C–S cycling and climate change across the Early Permian Artinskian warming. Figshare. Dataset. <https://doi.org/10.6084/m9.figshare.28942166.v2> (2025).
89. Scotese, C. R. PALEOMAP. <http://www.scotese.com> (2002).
90. Liu, D. et al. Paleoenvironment evolution of the Permian Lucaogou Formation in the southern Junggar Basin, NW China. *Paleogeogr. Paleoclimatol. Paleoecol.* **603**, 111198 (2022).
91. Timmreck, C. Modeling the climatic effects of large explosive volcanic eruptions. *Wires. Clim. Chang* **3**, 545–564 (2012).

Acknowledgements

We thank Editor, Associate Editor, and reviewers for constructive comments in improving the manuscript. This work was jointly funded by the National Key Research and Development Program of China (2024YFF0808100), the National Natural Science Foundation of China (Grant No. 42230808), and the National Science and Technology Major Project of China for Oil and Gas (2025ZD1400300). No permissions were required for the sampling in this study.

Author contributions

A.B.W., and J.C. designed and conceived the study; A.B.W., J.K.Z., Y.C.W., C.J.Z., and W.J.H. collected the samples; W.X.H. provided resources; A.B.W., and J.K.Z. contributed to the data curation; A.B.W., and J.C. contributed to writing and editing the original draft.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at

<https://doi.org/10.1038/s43247-025-02843-8>.

Correspondence and requests for materials should be addressed to Jian Cao.

Peer review information *Communications Earth and Environment* thanks Christopher R Fielding, Jianghai Yang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Jun Shen and Alireza Bahadori. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025